CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First / Second Semester of B. Tech. Examination (CL/ME/EE/CE/IT/EC)
May 2014

CL 101 / CL 101.01 Fundamentals of Civil Engineering

Date: 21.05.2014, Wednesday   Time: 01:30 p.m. To 04:30 p.m.   Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Discuss suitability and requirements of different modes of transportation. [07]

Q - 2 (a) Enlist branches of civil engineering and explain any two. [04]

(b) Attempt any two [08]

(i) Explain the fundamental principles of Surveying.

(ii) Explain reciprocal ranging.

(iii) A 20 m chain was found to be 8 cm too long at the end of a day's work after measuring 4000 m. If the chain was correct before commencement of work, find the correct length of the line.

(c) Define any two [02]

(i) Base line (ii) Surveying (iii) Tie Station

Q - 3 (a) Convert the following [05]

1) WCB to RB

(i) 285° (ii) 198°30' (iii) 58°

2) QB to WCB

(i) N23°W (ii) S82°E

(b) The following staff readings were recorded in a leveling operation. [05]

1.175, 2.605, 1.835, 2.405, 1.145, 0.965, 1.115, 1.785, 1.225, 1.645 & 1.715. Station A is the benchmark with reduced level 175.68m. Find the RLs of all the other points by rise and fall method. The first reading was taken on point A and instrument was shifted after the 2^{nd}, 6^{th} and 9^{th} readings. Apply the usual checks.

(c) Differentiate between Prismatic compass and Surveyor's compass. [04]

OR

Q - 3 (a) Explain the procedure to find area of irregular figure by Planimeter. [04]

(b) The following offsets were taken at 10m intervals from a survey line to an irregular boundary line. [05]

2.75, 4.50, 6.70, 5.30, 7.30, 8.50, 7.90, 6.30, 4.20, 3.50m

Calculate the area enclosed between the survey line, the irregular boundary line and the first and last offsets by The trapezoidal rule and Simpson's rule.

(c) The details of observed bearings are mentioned below. Find out the included angles and also correct the angles if needed to be corrected. [05]

Line	Fore Bearing	Back Bearing
AB	$20^0 30'$	$200^0 00'$
BC	$110^0 00'$	$290^0 30'$
CD	$195^0 00'$	$15^0 00'$
DA	$286^0 30'$	$106^0 00'$

SECTION – II

Q - 4 (a) i. Enlist different uses of cement. [02]

ii. Enlist properties of fresh Concrete. Write down the compressive strength of 53 grade cement at the end of 28 days. [02]

iii. Describe the characteristics of good bricks. [03]

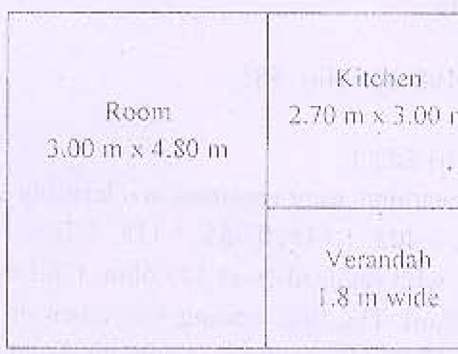
Q - 5 (a) i. Write down difference between Load Bearing and Framed Structure. [05]

ii. Enlist Classification of Building based on Occupancy. [02]

(b) Draw the chart showing classification of foundation and draw a sketch typical wall footing for 20 cm thick wall. [07]

OR

Q - 5 (a) Draw a detailed plan of a building from the line diagram. Assume suitable scale. Consider plinth height as 60 cm. Also prepare a schedule giving relevant description of doors and windows. All walls are 30 cm thick. [07]



(b) Enumerate various principles of planning and explain roominess and privacy in detail. [07]

Q - 6 (a) Write a short note on Setback and Built up area [06]

(b) Explain with a neat sketch the hydrological cycle. List various sources of water. [08]

OR

Q - 6 (a) Write down functions of the following [06]

(a) Plinth (b) Sill (c) Beam (d) Roof (e) Weather shed (f) Door

(b) Explain water conservation with its importance. [08]

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